

LECTURE SHEET

EXERCISE-I

*Problems on slopes, Equation of lines and Intercept form,
problems on areas, Normal form, symmetric form*

LEVEL-I (MAIN)

Single answer type questions

- The medians AD and BE of the triangle with vertices $A(0, b)$, $B(0, 0)$ and $C(a, 0)$ are mutually perpendicular if
 1) $b = \sqrt{2}a$ 2) $a = \sqrt{2}b$ 3) $b = -\sqrt{2}a$ 4) $a = 5\sqrt{2}b$
- The lines $p(p^2 + 1)x - y + q = 0$ and $(p^2 + 1)^2x + (p^2 + 1)y + 2q = 0$ are perpendicular to a common line for
 1) exactly one value of p 2) exactly two values of p
 3) more than two values of p 4) no values of p
- If PS is the median of the triangle with vertices $P(2, 2)$, $Q(6, -1)$ and $R(7, 3)$ then the equation of the line passing through $(1, -1)$ and parallel to $\overline{PS}$
 1) $4x - 7y - 11 = 0$ 2) $2x + 9y + 7 = 0$ 3) $4x + 7y + 3 = 0$ 4) $2x - 9y - 11 = 0$
- The perpendicular bisector of the line segment joining $P(1, 4)$ and $Q(k, 3)$ has y -intercept equal to -4 . Then a possible value of k is
 1) 1 2) 2 3) -2 4) -4
- The number of integral points (integral point means both the coordinates should be integers) exactly in the interior of the triangle with vertices $(0, 0)$, $(0, 21)$ and $(21, 0)$ is
 1) 133 2) 190 3) 233 4) 105
- The sum of the intercepts cut off by the axes on lines $x + y = a$, $x + y = ar$, $x + y = ar^2$, where $a \neq 0$ and $r = \frac{1}{2}$
 1) $2a$ 2) $a\sqrt{2}$ 3) $2\sqrt{2}a$ 4) a
- The locus of the midpoint of the portion between the axes of $x \cos \alpha + y \sin \alpha = p$, where p is a constant, is
 1) $x^2 + y^2 = \frac{4}{p^2}$ 2) $x^2 + y^2 = 4$ 3) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{2}{p^2}$ 4) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{4}{p^2}$
- A straight line through the point $A(3, 4)$ is such that its intercept between the axes is bisected at A . Its equation is
 1) $x + y = 7$ 2) $3x - 4y + 7 = 0$ 3) $4x + 3y = 24$ 4) $3x + 4y = 25$
- A line is drawn through the point $(1, 2)$ to meet the coordinate axes at P and Q such that it forms a triangle OPQ , where O is the origin. If the area of the triangle OPQ is least, then the slope of the line PQ is:
 1) $-1/4$ 2) -4 3) -2 4) $-1/2$

10. Let $O(0, 0)$, $P(3, 4)$, $Q(6, 0)$ be the vertices of the triangle OPQ . The point R inside the triangle OPQ is such that the triangles OPR , PQR , OQR are of equal area. The coordinates of R are
- 1) $\left(\frac{4}{3}, 3\right)$ 2) $\left(3, \frac{2}{3}\right)$ 3) $\left(3, \frac{4}{3}\right)$ 4) $\left(\frac{4}{3}, \frac{2}{3}\right)$
11. A straight line passing through $(2, 2)$ and cuts the triangle of area 9 sq. units from the first quadrant. The sum of all possible values for the slopes of such lines is
- 1) -2.5 2) -2 3) -1.5 4) -1
12. Area bounded by the lines $|2x-3| + |3y+4| = 5$ is
- 1) $\frac{25}{3}$ 2) $\frac{20}{3}$ 3) $\frac{17}{3}$ 4) $\frac{10}{3}$
13. A straight line makes an angle $\frac{\pi}{3}$ with positive direction of x -axis measured in anti clock direction and makes positive intercept on y -axis and the line is at a distance of 5 units from origin then its equation is
- 1) $y = \sqrt{3}x - 10$ 2) $y = \sqrt{3}x + 10$ 3) $y = \sqrt{3}x + 8$ 4) $y = \sqrt{3}x - 8$
14. The equation of a line passing through the point $(3, 2)$ and cutting an intercept of 2 units between the lines $3x + 4y = 11$ and $3x + 4y = 1$ is
- 1) $2x + y - 8 = 0$ 2) $3y - 4x + 6 = 0$ 3) $3x + 4y - 17 = 0$ 4) $2x - y - 4 = 0$
15. If p and q are the perpendicular distance from the origin to the straight lines $x \sec \theta - y \csc \theta = a$ and $x \cos \theta + y \sin \theta = a \cos 2\theta$ then
- 1) $4p^2 + q^2 = a^2$ 2) $p^2 + q^2 = a^2$ 3) $p^2 + 2q^2 = a^2$ 4) $4p^2 + q^2 = 2a^2$
16. $A(0, 2)$ and $C(6, 4)$ are the opposite vertices of a square $ABCD$. Sum of the slopes of sides through vertex A will be
- 1) $\frac{1}{2}$ 2) $\frac{5}{2}$ 3) $\frac{9}{2}$ 4) $\frac{3}{2}$

LEVEL-II (ADVANCED)

Single answer type questions

1. The line $2x + 3y = 6$, $2x + 3y = 8$ cut the X -axis at A, B respectively. A line l drawn through the point $(2, 2)$ meets the X -axis at C in such a way that abscissae of A, B and C are in arithmetic progression. Then the equation of the line l is
- a) $2x + 3y = 10$ b) $3x + 2y = 10$ c) $2x - 3y = 10$ d) $3x - 2y = 10$
2. The area of the triangle formed by the axes and the line $(\cosh \alpha - \sinh \alpha)x + (\cosh \alpha + \sinh \alpha)y = 2$ in sq. unit, is
- a) 4 b) 3 c) 2 d) 1
3. If a straight line L is perpendicular to the line $4x - 2y = 1$ and forms a triangle of area 4 square units with the coordinate axes, then an equation of the line L is
- a) $2x + 4y + 7 = 0$ b) $2x - 4y + 8 = 0$ c) $2x + 4y + 8 = 0$ d) $4x - 2y - 8 = 0$

4. $OPQR$ is a square and M, N are the middle points of the sides PQ and QR respectively then the ratio of the area of the square and the triangle OMN is
 a) 4:1 b) 2:1 c) 8:3 d) 4:3
5. If $f(x+y) = f(x)f(y)$ for all x and y if $f(1) = 2$, then area enclosed by $3|x| + 2|y| \leq 8$ is
 a) $f(5)$ sq. units b) $f(6)$ sq. units c) $\frac{1}{3} f(6)$ sq. units d) $f(4)$ sq. units
6. $P(m, n)$ (where m, n are natural numbers) is any point in the interior of the quadrilateral formed by the pair of lines $xy = 0$ and the lines $2x + y - 2 = 0$ and $4x + 5y = 20$. The possible number of positions of the point P is
 a) 7 b) 5 c) 4 d) 6
7. AB is a variable line sliding between the coordinate axes in such a way that A lies on x -axis and B lies on y -axis. If P is a variable point on AB such that $PA = b$, $PB = a$ and $AB = a + b$, then the equation of the locus of P is
 a) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ b) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ c) $x^2 + y^2$ d) $a^2 + b^2$
8. A line has intercepts a and b on the coordinate axes. When the axes are rotated through an angle α , keeping the origin fixed, the line makes equal intercepts on the coordinate axes, then $\tan \alpha =$
 a) $\frac{a+b}{a-b}$ b) $\frac{a-b}{a+b}$ c) $a^2 - b^2$ d) $a^2 + b^2$

More than one correct answer type questions

9. The coordinates of the points A and B are respectively $(-3, 2)$ and $(2, 3)$. P and Q are points on the line joining A and B such that $AP = PQ = QB$. A square $PQRS$ is constructed on PQ as one side, the coordinates of R can be
 a) $\left(-\frac{4}{3}, \frac{7}{3}\right)$ b) $\left(0, \frac{13}{3}\right)$ c) $\left(\frac{1}{3}, \frac{8}{3}\right)$ d) $\left(\frac{2}{3}, 1\right)$
10. If the straight line $ax + cy = 2b$, where $a, b, c > 0$, makes a triangle of area 2 sq. units with the coordinate axes, then
 a) a, b, c are in GP b) $a, -b, c$ are in GP c) $a, 2b, c$ are in GP d) $a, -2b, c$ are in GP
11. A line through $A(-5, -4)$ with slope $\tan \theta$ meets the lines $x + 3y + 2 = 0$, $2x + y + 4 = 0$, $x - y - 5 = 0$ at B, C, D respectively, such that $\left(\frac{15}{AB}\right)^2 + \left(\frac{10}{AC}\right)^2 = \left(\frac{6}{AD}\right)^2$ then
 a) $\frac{15}{AB} = \cos \theta + 3 \sin \theta$ b) $\frac{10}{AC} = 2 \cos \theta + \sin \theta$
 c) $\frac{6}{AD} = \cos \theta - \sin \theta$ d) Slope of the line is $-\frac{2}{3}$
12. Let $B(1, -3)$ and $D(0, 4)$ represents two vertices of a rhombus $ABCD$ in xy plane, then coordinates of vertex A if $\angle BAD = 60^\circ$ can be equal to:
 a) $\left(\frac{1-7\sqrt{3}}{2}, \frac{1-\sqrt{3}}{2}\right)$ b) $\left(\frac{1+7\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}\right)$ c) $\left(\frac{-1+7\sqrt{3}}{2}, \frac{-1+\sqrt{3}}{2}\right)$ d) $\left(\frac{-1-7\sqrt{3}}{2}, \frac{-1-\sqrt{3}}{2}\right)$

Linked comprehension type questions

Passage - I :

$$L_1 : 3x + 4y - 8 = 0 ; L_2 : 2x + 7y - 1 = 0$$

13. If L_1, L_2 represents the sides AB and AC of the isosceles triangle ABC with $AB = AC = 2$ then the coordinates of

a) B are $(28/5, -11/5)$

b) B are $(28/15, 1/5)$

c) C are $\left(\frac{14+4\sqrt{53}}{\sqrt{53}}, \frac{4+\sqrt{5}}{\sqrt{5}}\right)$

d) C are $\left(\frac{14+4\sqrt{53}}{\sqrt{53}}, \frac{4-\sqrt{5}}{\sqrt{5}}\right)$

14. Equation of the line through B parallel to AC is

a) $2x + 7y + 21 = 0$

b) $10x + 35y = 63$

c) $10x + 35y + 21 = 0$

d) $2x + 7y = 63$

15. If D is the midpoint of BC and E is the mid point of CA then DE is equal to

a) $1/4$

b) $1/2$

c) 1

d) 12

Integer answer type questions

16. The line $3x + 2y = 24$ meets the y -axis at A and the x -axis at B . The perpendicular bisector of AB meets the line through $(0, -1)$ parallel to the x -axis at C . If the area of triangle ABC is A , then the Value of $A/13$ is _____
17. The number of possible straight lines passing through $(2, 3)$ and forming a triangle with co-ordinate axes whose area is 12 sq. units is
18. A straight line passing through the point $(87, 33)$ cuts the positive direction of the co-ordinate axes at the point P and Q . If O is the origin then minimum area of the triangle OPQ is P then $P - 5740$ is

EXERCISE-II

(Problems on distances, Ratios, problems on point of intersection, concurrent lines)

LEVEL-I (MAIN)Single answer type questions

1. If the base of an isosceles triangle is of length $2p$ and the length of the altitude dropped to the base is q , then the distance from the mid point of the base to the side of the triangle is

1) $\frac{pq}{\sqrt{p^2 + q^2}}$

2) $\frac{2pq}{\sqrt{p^2 + q^2}}$

3) $\frac{3pq}{\sqrt{p^2 + q^2}}$

4) $\frac{4pq}{\sqrt{p^2 + q^2}}$

2. An equilateral triangle is constructed between two parallel lines $\sqrt{3}x + y - 6 = 0$ and $\sqrt{3}x + y + 9 = 0$ with base on one line and vertex on the other. Then the area of triangle is

1) $\frac{200}{\sqrt{3}}$

2) $\frac{225}{4\sqrt{3}}$

3) $\frac{225}{\sqrt{3}}$

4) $\frac{200}{4\sqrt{3}}$

3. PQR is an equilateral triangle such that the vertices Q and R lie on the lines $x + y = \sqrt{2}$ and $x + y = 7\sqrt{2}$ respectively. If P lies between the two lines at a distance 4 from one of them then the length of side of equilateral triangle PQR is (in units)
- 1) 8 2) $\frac{4\sqrt{7}}{\sqrt{3}}$ 3) $\frac{\sqrt{85}}{3}$ 4) $\frac{4\sqrt{5}}{\sqrt{3}}$
4. The distance of the point $(2, 3)$ from the line $2x - 3y + 9 = 0$ measured along a line $x - y + 1 = 0$ is
- 1) $4\sqrt{2}$ 2) $2\sqrt{2}$ 3) $\sqrt{2}$ 4) $\frac{1}{\sqrt{2}}$
5. If p_1, p_2 are the perpendicular distances from the origin to the two straight lines which are perpendicular to each other, then the locus of the point of intersection of the perpendicular lines is
- 1) $x^2 + y^2 = p_1 + p_2$ 2) $x^2 + y^2 = p_1^2 + p_2^2$ 3) $\frac{x^2}{p_1^2} + \frac{y^2}{p_2^2} = 1$ 4) $xy = p_1 p_2$
6. If the points $(1, 2)$ and $(3, 4)$ were to be on the same side of the line $3x - 5y + a = 0$ then
- 1) $7 < a < 11$ 2) $a = 7$ 3) $a = 1$ 4) $a < 7$ (or) $a > 11$
7. If the line $2x + y = k$ passes through the point which divides the line segment joining the points $(1, 1)$ and $(2, 4)$ in the ratio $3 : 2$, then k equals
- 1) $\frac{29}{5}$ 2) 5 3) 6 4) $\frac{11}{5}$
8. A straight line through the origin O meets the parallel lines $4x + 2y = 9$ and $2x + y + 6 = 0$ at points P and Q respectively. Then the point O divides the segment PQ in the ratio
- 1) $1 : 2$ 2) $3 : 4$ 3) $2 : 1$ 4) $4 : 3$
9. $P(m, n) (m, n \in \mathbb{N})$ is any point in the interior of the quadrilateral formed by the pair of lines $xy = 0$ and two lines $2x + y - 2 = 0$ and $4x + 5y = 20$ then the possible number of positions of the points P is
- 1) 4 2) 5 3) 6 4) 11
10. A line L cuts the sides AB, BC of $\triangle ABC$ in the ratio $2 : 5, 7 : 4$ respectively then the line L cuts CA in the ratio
- 1) $7 : 10$ 2) $2 : -5$ 3) $10 : -7$ 4) $5 : -2$
11. Let ABC be a triangle. If P is a point such that AP divides BC in the ratio $2 : 3, BP$ divides CA in the ratio $3 : 5$ then the ratio in which CP divides AB is
- 1) $2 : 5$ 2) $2 : -5$ 3) $5 : 2$ 4) $5 : -2$
12. If the point (a, a) falls between the lines $|x + y| = 2$, then:
- 1) $|a| = 2$ 2) $|a| = 1$ 3) $|a| < 1$ 4) $|a| < \frac{1}{2}$
13. If $(\sin\theta, \cos\theta), \theta \in [0, 20]$ and $(1, 4)$ lie on the same side or on the line $\sqrt{3}x - y + 1 = 0$, then the maximum value of $\sin\theta$ will be
- 1) 1 2) 0 3) $\frac{1}{2}$ 4) $1 - \frac{\sqrt{3}}{2}$

14. The equation of the line passing through the point of intersection of the lines $3x + 2y + 4 = 0$, $2x + 5y - 1 = 0$ and whose distance from the point $(2, -1)$ is 2 is
- 1) $y + 1 = 0$, $4x - 3y - 5 = 0$ 2) $y - 1 = 0$, $4x + 3y + 5 = 0$
 3) $y + 1 = 0$, $4x - 3y + 5 = 0$ 4) $y - 1 = 0$, $4x + 3y - 5 = 0$
15. Given the points $A(6, 3)$, $B(-3, 5)$ and $C(4, -2)$. If P is any point (x, y) then find the ratio of the areas of triangle PBC and triangle ABC is
- 1) $|x+y-2| : 7$ 2) $|x+y+2| : 7$ 3) $|x-y+2| : 7$ 4) $|x-y-2| : 7$
16. If θ is the parameter, then the family of lines represented by $(2\cos\theta + 3\sin\theta)x + (3\cos\theta - 5\sin\theta)y - (5\cos\theta - 7\sin\theta) = 0$ are concurrent at the point
- 1) $(-1, 1)$ 2) $(-1, -1)$ 3) $(1, 1)$ 4) $\left(\frac{4}{19}, \frac{29}{19}\right)$
17. Equation of the line belonging to the system $3x + 2y - 8 + \lambda(2x - 5y + 1) = 0$ and having equal intercepts is
- 1) $x + y = 0$ 2) $x + y = 1$ 3) $x + y = 2$ 4) $x + y = 3$
18. The ratio in which the line $3x + 4y + 2 = 0$ divides the distance between $3x + 4y + 5 = 0$ and $3x + 4y - 5 = 0$ is
- 1) $7 : 3$ 2) $3 : 7$ 3) $2 : 3$ 4) $3 : 2$

LEVEL-II (ADVANCED)

Single answer type questions

1. The value of k such that the lines $2x - 3y + k = 0$, $3x - 4y - 13 = 0$ and $8x - 11y - 33 = 0$ are concurrent, is
- a) 20 b) -7 c) 7 d) -20
2. The equation of the line with gradient $-3/2$, which is concurrent with the lines $4x + 3y - 7 = 0$ and $8x + 5y - 1 = 0$, is
- a) $3x + 2y - 2 = 0$ b) $3x + 2y - 63 = 0$
 c) $2y - 3x - 2 = 0$ d) $2y - 3x - 63 = 0$
3. Let a , b , c and d be non zero numbers. If the point of intersection of the lines $4ax + 2ay + c = 0$ and $5bx + 2by + d = 0$ lies in the fourth quadrant and is equidistant from the two axes then
- a) $2bc - 3ad = 0$ b) $2bc + 3ad = 0$
 c) $3bc - 2ad = 0$ d) $3bc + 2ad = 0$
4. Two sides of a rhombus are along the lines, $x - y + 1 = 0$ and $7x - y - 5 = 0$. If its diagonals intersect at $(-1, -2)$, then which one of the following is a vertex of this rhombus?
- a) $(-3, -8)$ b) $\left(\frac{1}{3}, -\frac{8}{3}\right)$ c) $\left(-\frac{10}{3}, -\frac{7}{3}\right)$ d) $(-3, -9)$

5. The line parallel to the x -axis and passing through the intersection of the lines $ax + 2by + 3b = 0$ and $bx - 2ay - 3a = 0$, where $(a, b) \neq (0, 0)$, is
- a) below the x -axis at a distance of $3/2$ from it b) below the x -axis at a distance of $2/3$ from it
c) below the x -axis at a distance of $2/3$ from it d) above the x -axis at a distance of $2/3$ from it.
6. If the lines $x + ay + a = 0$, $bx + y + b = 0$, $cx + cy + 1 = 0$ (a, b, c being distinct and $\neq 1$) are concurrent then the value of $\frac{a}{a-1} + \frac{b}{b-1} + \frac{c}{c-1} =$
- a) -1 b) 0 c) 1 d) 3
7. If $5a + 4b + 20c = t$, then the value of t for which the line $ax + by + c - 1 = 0$ always passes through a fixed point is
- a) 0 b) 20 c) 30 d) None of these
8. If non-zero numbers a, b, c are H.P., then the straight line $\frac{x}{a} + \frac{y}{b} + \frac{1}{c} = 0$ always passes through a fixed point. That point is
- a) $(-1, 2)$ b) $(-1, -2)$ c) $(1, -2)$ d) $(1, -2/2)$
9. The number of lines that can be drawn through the point $(4, -5)$ at a distance of 10 units from the point $(1, 3)$ is
- a) 0 b) 1 c) 2 d) Infinite
10. The number of circles that touch all the 3 lines $2x + y = 3$, $4x - y = 3$, $x + y = 2$ is
- a) 0 b) 1 c) 2 d) 4
11. For $a > b > c > 0$, the distance between $(1, 1)$ and the point of intersection of the lines $ax + by + c = 0$ and $bx + ay + c = 0$ less than $2\sqrt{2}$. Then and
- a) $a + b - c > 0$ b) $a - b + c < 0$ c) $a - b + c > 0$ d) $a + b - c < 0$
12. If $\frac{a}{\sqrt{bc}} - 2 = \sqrt{\frac{b}{c}} + \sqrt{\frac{c}{b}}$ where $a, b, c > 0$ then family of lines $\sqrt{a}x + \sqrt{b}y + \sqrt{c} = 0$ passes through the point
- a) $(1, 1)$ b) $(1, -2)$ c) $(-1, 2)$ d) $(-1, 1)$
13. A variable line drawn through the point of intersection of the lines $\frac{x}{a} + \frac{y}{b} = 1$, $\frac{x}{b} + \frac{y}{a} = 1$ meets the coordinate axes in A and B . Then the locus of midpoint of AB is
- a) $2xy(a + b) = ab(x + y)$ b) $xy(a + b) = ab(x + y)$ c) $2xy(a + b) = ab(x - y)$ d) $xy(a + b) = ab(x - y)$

More than one correct answer type questions

14. If P is a point (x, y) on the line $y = -3x$ such that P and the point $(3, 4)$ are on the opposite sides of the line $3x - 4y = 8$, then
- a) $x > 8/15$ b) $x > 8/5$ c) $y < -8/5$ d) $y < -8/15$
15. If $6a^2 + 12b^2 + 2c^2 + 17ab - 10bc - 7ac = 0$ then all the lines represented by $ax + by + c = 0$ are concurrent at the point
- a) $(-2, -3)$ b) $(2, 3)$ c) $\left(-\frac{3}{2}, -2\right)$ d) $\left(-\frac{3}{2}, 2\right)$

16. If the Points $\left(\frac{a^3}{(a-1)}, \frac{(a^2-3)}{(a-1)}\right)$, $\left(\frac{b^3}{(b-1)}, \frac{(b^2-3)}{(b-1)}\right)$, and $\left(\frac{c^3}{(c-1)}, \frac{(c^2-3)}{(c-1)}\right)$, where a, b, c are different from 1, lie on the line $lx + my + n = 0$ then
- a) $a+b+c = -\frac{m}{l}$
- b) $ab+bc+ca = \frac{n}{l}$
- c) $abc = \frac{(m+n)}{l}$
- d) $abc-(bc+ca+ab)+3(a+b+c)=0$
17. Two lines from the family of lines $(1+2\lambda)x - (1+\lambda)y + 1 = 0$ and the line $x+y=5$ form an equilateral triangle. The equation of the two lines can be
- a) $y-2=(2+\sqrt{3})(x-1)$
- b) $y-2=(2-\sqrt{3})(x-1)$
- c) $y-2=(3-\sqrt{2})(x-1)$
- d) $y-1=(3-\sqrt{2})(x-1)$

Linked comprehension type questions

Passage - I

Let L be the line belonging to the family of the straight lines $(a+2b)x + (a-3b)y + a-8b = 0$, $a, b \in \mathbb{R}$, which is farthest from the point $(2, 2)$

18. The equation of line L is
a) $x + 4y + 7 = 0$ b) $2x + 3y + 4 = 0$ c) $4x - y - 6 = 0$ d) none of these
19. Area formed by the line L with coordinate axis is
a) $4/3$ sq.units b) $9/2$ sq units c) $49/8$ sq.units d) none of these
20. If L is concurrent with the lines $x - 2y + 1 = 0$ and $3x - 4y + \lambda = 0$, then the value of λ is
a) 2 b) 1 c) -4 d) 5

Matrix matching type questions

- | | | |
|-----|--|--------------------|
| 21. | Column - I | Column - II |
| A) | An integer value of m for which the x -coordinate of the point of intersection of the lines $3x + 4y = 9$ and $y = mx + 1$ is also an integer is | p) $m = -1$ |
| B) | The line $2x - 3y - 6 = 0$ meets the coordinate axes at $(a, 0)$ and $(0, b)$. If the line $y = mx + m^2$ passes through (a, b) , then | q) $m = -2$ |
| C) | If the point $(3, 4)$ lies on the locus of the point of intersection of the lines $x \cos \alpha + y \sin \alpha = m$ and $x \sin \alpha - y \cos \alpha = n$ such that $3m - 4n = 0$, then | r) $m = 4$ |
| D) | If the line $y = mx - 8/m$ meets the curve $y^2 = 4x$ at a point where the curve meets the line $y = 2$, then | s) $m = -4$ |

Integer answer type questions

22. Two rays in the first quadrant $x + y = |a|$ and $ax - y = 1$ intersect each other in the interval $a \in (a_0, \infty)$, the value of a_0 is

23. For all real values of a and b , lines $(2a+b)x + (a+3b)y + (b-3a) = 0$ and $mx + 2y + 6 = 0$ are concurrent. Then $|m|$ is equal to
24. Let the algebraic sum of the perpendicular distances from the points $(2, 0)$, $(0, 2)$ and $(1, 1)$ to a variable straight line be zero; then the line passes through a fixed point whose coordinates are (a, b) , then $a + b$ is

EXERCISE-III

(Angle between lines, foot of the perpendicular, Image, centres of triangle)

LEVEL-I (MAIN)

Single answer type questions

1. The diagonals of the parallelogram whose sides are $lx + my + n = 0$, $lx + my + n' = 0$, $mx + ly + n = 0$, $mx + ly + n' = 0$ include an angle
- 1) $\pi/3$ 2) $\pi/2$ 3) $\tan^{-1}\left(\frac{l^2 - m^2}{l^2 + m^2}\right)$ 4) $\tan^{-1}\left(\frac{2lm}{l^2 + m^2}\right)$
2. If (a, a^2) falls inside the angle made by the lines $y = \frac{x}{2}$, $x > 0$, and $y = 3x$, $x > 0$, then a belongs to
- 1) $\left(0, \frac{1}{2}\right)$ 2) $(3, \infty)$ 3) $\left(\frac{1}{2}, 3\right)$ 4) $\left(-3, -\frac{1}{2}\right)$
3. A straight line L through the point $(3, -2)$ is inclined at an angle 60° to the line $\sqrt{3}x + y = 1$. If L also intersects the x -axis, then the equation of L is
- 1) $y + \sqrt{3}x + 2 - 3\sqrt{3} = 0$ 2) $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$
 3) $\sqrt{3}y - x + 3 + 2\sqrt{3} = 0$ 4) $\sqrt{3}y + x - 3 + 2\sqrt{3} = 0$
4. The image of the line $x + 2y = 5$ in the line $x - y = 2$ is
- 1) $2x + y = 7$ 2) $2x - y = 7$ 3) $x + y = 2$ 4) $x - 2y = 7$
5. A and B are two variable points on x, y axes respectively such that $OA + OB = c$. The locus of the foot of the perpendicular from origin on this line $\overline{AB}$ is
- 1) $x^2 + y^2 = cxy$ 2) $x^2 + y^2 = 2cxy$
 3) $(x + y)(x^2 + y^2) = cxy$ 4) $(x + y)(x^2 + y^2) = 2cxy$
6. The equation of perpendicular bisectors of sides AB, BC of $\triangle ABC$ are $x - y - 5 = 0$, $x + 2y = 0$ respectively and $A(1, -2)$ then coordinate of C are
- 1) $(1, 0)$ 2) $(0, 1)$ 3) $(5, 0)$ 4) $(0, 0)$
7. $(2, 3)$ is a point on the side AB of $\triangle ABC$. The third vertex C moves such that the sides AC, BC are bisected by $x^2 - y^2 = 0$ at right angles. Then C lies on
- 1) $2x - 3y = 0$ 2) $3x - 2y = 0$ 3) $2x + 3y = 0$ 4) $3x + 2y = 0$

8. The equation of line segment AB is $y = x$. If A & B lie on same side of line mirror $2x - y = 1$, then the equation of image of AB with respect to line mirror $2x - y = 1$ is
- 1) $y = 7x - 5$ 2) $y = 7x - 6$ 3) $y = 3x - 7$ 4) $y = 6x - 5$
9. L_1 and L_2 are two intersecting lines. If the image of L_1 w.r.t. L_2 and that of L_2 w.r.t. L_1 coincide, then the angle between L_1 and L_2 is
- 1) 35° 2) 60° 3) 90° 4) 45°
10. If $2x + y - 4 = 0$ is bisector of the angle between the lines $a(x-1) + b(y-2) = 0$, $c(x-1) + d(y-2) = 0$, then the other bisector is
- 1) $x - 2y + 1 = 0$ 2) $x - 2y - 3 = 0$ 3) $x - 2y + 3 = 0$ 4) $x - 2y - 5 = 0$
11. Point $(2, 4)$ is translated through a distance $3\sqrt{2}$ units measured parallel to the line $y - x = 1$, in the direction of decreasing ordinates, to reach at Q . If R is the image of Q with respect to the line $y - x = 1$, then coordinates of R are
- 1) $(0, 0)$ 2) $(-1, 1)$ 3) $(6, 6)$ 4) $(5, 7)$
12. The point B is the image of A in the line $x + y + 4 = 0$ and C is the image of B in the line $2x - y + 2 = 0$. If $A = (1, 2)$ then circum diameter of triangle ABC is
- 1) 12 2) 10 3) 8 4) 5
13. Let $P = (-1, 0)$, $Q = (0, 0)$, and $R = (3, 3\sqrt{3})$ be three points. Then the equation of the bisector of $\angle PQR$ is
- 1) $(\sqrt{3}/2)x + y = 0$ 2) $x + \sqrt{3}y = 0$
 3) $\sqrt{3}x + y = 0$ 4) $x + (\sqrt{3}/2)y = 0$
14. Lines $L_1 : y - x = 0$ and $L_2 : 2x + y = 0$ intersect the line $L_3 : y + 2 = 0$ at P and Q , respectively. The bisector of the acute angle between L_1 and L_2 intersects L_3 at R .
- Statement-1 :** The ratio $PR : RQ$ equals $2\sqrt{2} : \sqrt{5}$. Because
- Statement-2 :** In any triangle, bisector of an angle divides the triangle into two similar triangles.
- 1) Statement 1 is correct and statement 2 is correct explanation of statement 1
 2) Statement 1 is correct and statement 2 is not correct explanation of statement 1
 3) Statement 1 is correct statement 2 is false
 4) Statement 1 is false statement 2 is correct
15. Two sides of a triangle are along the coordinate axes and the medians through the vertices (other than the origin) are mutually perpendicular. The number of such triangles is/are
- 1) zero 2) two 3) four 4) infinite
16. The vertices of a triangle lie on the lines $y = x \tan \theta_1$, $y = x \tan \theta_2$, $y = x \tan \theta_3$. The circumcentre is $(0, 0)$, then the locus of orthocentre is
- 1) $\sum (\sin \theta_i)x + \sum (\cos \theta_i)y = 0$ 2) $\sum (\sin \theta_i)x - \sum (\cos \theta_i)y = 0$
 3) $x + y = 0$ 4) $x - y = 0$

17. The locus of the orthocenter of the triangle formed by the lines $(1+p)x - py + p(1+p) = 0$, $(1+q)x - qy + q(1+q) = 0$ and $y = 0$, where $p \neq q$, is
- 1) $x+y = pq$ 2) $x+y = q$ 3) $x+y = p$ 4) $x+y = 0$
18. If the ends of the base of an isosceles triangle are at $(2, 0)$ and $(0, 1)$, and the equation of one side is $x = 2$, then the orthocenter of the triangle is
- 1) $(3/2, 3/2)$ 2) $(5/4, 1)$ 3) $(3/4, 1)$ 4) $(4/3, 7/12)$

LEVEL-II (ADVANCED)

Single answer type questions

1. Vertex A of $\triangle ABC$ moves in such way that $\tan B + \tan C = \text{constant}$, when BC is fixed then locus of orthocentre of $\triangle ABC$ is a
- a) Straight line b) St.line parallel to $y = x$
c) St.line parallel to $y = -x$ d) locus not found
2. A and B are two fixed points, the vertex C and $\triangle ABC$ moves such that $\cot A + \cot B$ is constant. Then locus of C is
- a) a line perpendicular to AB b) a line parallel to AB
c) a parabola d) not a straight line
3. A straight line L through the origin meets the lines $x + y = 1$ and $x + y = 3$ at P and Q , respectively. Lines L_1 and L_2 are drawn, parallel to $2x - y = 5$ and $3x + y = 5$ are passing through P , Q respectively. Lines L_1 and L_2 intersect at R . Then that the locus of R , as L varies, is a straight line.
- a) $x - 3y + 5 = 0$ b) $x + 3y + 5 = 0$ c) $x + 3y - 5 = 0$ d) $x - 3y - 5 = 0$
4. The area of the triangle formed by the intersection of a line parallel to the x -axis and passing through $P(h, k)$ with the lines $y = x$ and $x + y = 2$ is $4h^2$. Then the locus of the point P is.
- a) $y = 2x + 1$ b) $y = 3x + 1$ c) $y = 2x - 1$ d) $y = 2x + 2$
5. A straight line L through the point $(3, -2)$ is inclined at an angle 60° to the line $\sqrt{3}x + y = 1$, if L also intersects the x -axis, then the equation of L is
- a) $y + \sqrt{3}x + 2 - 3\sqrt{3} = 0$ b) $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$
c) $\sqrt{3}y - x + 3 + 2\sqrt{3} = 0$ d) $\sqrt{3}y + x - 3 + 2\sqrt{3} = 0$
6. Locus of the image of the point $(2, 3)$ in the line $(2x - 3y + 4) + k(x - 2y + 3) = 0, k \in R$, is a
- a) straight line parallel to x -axis b) straight line parallel to y -axis
c) circle of radius $\sqrt{2}$ d) circle of radius 3
7. In $\triangle ABC$, equation of BC is $x - y = 0$. If the circumcentre and orthocentre are $(2, 3)$, $(5, 8)$, then the circumradius is
- a) $\sqrt{10}$ b) $2\sqrt{10}$ c) $3\sqrt{10}$ d) $4\sqrt{10}$

8. The Incentre of a triangle formed by the lines $x \cos\left(\frac{\pi}{9}\right) + y \sin\left(\frac{\pi}{9}\right) = \pi$; $x \cos\left(\frac{8\pi}{9}\right) + y \sin\left(\frac{8\pi}{9}\right) = \pi$; $x \cos\left(\frac{13\pi}{9}\right) + y \sin\left(\frac{13\pi}{9}\right) = \pi$ is
- a) $\left(\cos\frac{\pi}{9}, \sin\frac{\pi}{9}\right)$ b) $\left(\cos\frac{8\pi}{9}, \sin\frac{8\pi}{9}\right)$
 c) (0,0) d) $\left(\cos\frac{13\pi}{9}, \sin\frac{13\pi}{9}\right)$
9. A ray of light along $x + \sqrt{3}y = \sqrt{3}$ gets reflected upon reaching x -axis, the equation of the reflected ray is:
- a) $y = x + \sqrt{3}$ b) $\sqrt{3}y = x - \sqrt{3}$
 c) $y = \sqrt{3}x - \sqrt{3}$ d) $\sqrt{3}y = x - 1$
10. If the lines $x+y=0$, $x-y=0$ and $2x+3y-6=0$ are the sides of a triangle and $(-2, a)$ is an interior point of the triangle then a lies
- a) $2 < a < 10/3$ b) $-2 < a < 10/3$
 c) $-1 < a < 9/2$ d) $-3 > a > 9/2$

More than one correct answer type questions

11. In $\triangle ABC$ $A(-1,-1)$, $B(4,5)$ and C lies on the straight line $y=5x-15$. If area of the $\triangle ABC$ is $\frac{19}{2}$ sq.units, then the point 'c' may be
- a) (5, 10) b) (3, 0) c) (2, -5) d) (5, 4)
12. Two straight lines $u = 0$ and $V = 0$ pass through the origin and the angle between them is $\tan^{-1}(7/9)$. If the ratio of the slope of $v = 0$ and $u = 0$ is $9/2$, then their equations are
- a) $y + 3x = 0$ and $3y + 2x = 0$ b) $2y + 3x = 0$ and $3y + x = 0$
 c) $2y = 3x$ and $3y = x$ d) $y = 3x$ and $3y = 2x$
13. Let $L_1 : 3x + 4y - 1 = 0$ and $L_2 : 5x - 12y + 2 = 0$ be two given lines. Let image of every point on L_1 w.r.t a line L lies on L_2 then possible equation of 'L' can be
- a) $14x + 112y - 23 = 0$ b) $64x - 8y - 3 = 0$ c) $11x - 4y = 0$ d) $52y - 45x = 7$
14. The equation of a straight line passing through the point (2,3) and inclined at an angle of $\tan^{-1}(1/2)$ with the line $y + 2x = 5$ is
- a) $y = 3$ b) $x = 2$ c) $3x + 4y - 18 = 0$ d) $4x + 3y - 17 = 0$

Linked comprehension type questions**Passage-I :**

$A(1, 3)$ and $C\left(-\frac{2}{5}, -\frac{2}{5}\right)$ are the vertices of a triangle ABC and the equation of the angle bisector of $\angle ABC$ is $x + y = 2$

15. Equation of side BC is

- a) $7x + 3y - 4 = 0$ b) $7x + 3y + 4 = 0$
 c) $7x - 3y + 4 = 0$ d) $7x - 3y - 4 = 0$

16. Coordinates of vertex B are

- a) $\left(\frac{3}{10}, \frac{17}{10}\right)$ b) $\left(\frac{17}{10}, \frac{3}{10}\right)$ c) $\left(-\frac{5}{2}, \frac{9}{2}\right)$ d) $(1, 1)$

17. Equation of side AB is

- a) $3x + 7y = 24$ b) $3x + 7y + 24 = 0$
 c) $13x + 7y + 8 = 0$ d) $13x - 7y + 8 = 0$

Passage - II :

Consider a variable line 'L' which passes through the point of intersection P of the lines $3x+4y-12=0$ and $x+2y-5=0$ meeting the coordinate axes at point A and B .

18. Locus of the middle point of the segment AB has the equation

- a) $3x + 4y = 4xy$ b) $3x + 4y = 3xy$ c) $4x + 3y = 4xy$ d) $4x + 3y = 3xy$

19. Locus of the feet of the perpendicular from the origin on the variable line L has the equation

- a) $2(x^2 + y^2) - 3x - 4y = 0$ b) $2(x^2 + y^2) - 4x - 3y = 0$
 c) $x^2 + y^2 - 3x - y = 0$ d) $x^2 + y^2 - x - 2y = 0$

20. Locus of the centroid of the variable triangle OAB has the equation (where O is origin)

- a) $3x + 4y + 6xy = 0$ b) $4x + 3y - 6xy = 0$
 c) $3x + 4y - 6xy = 0$ d) $4x + 3y + 6xy = 0$

Integer answer type questions

21. If the co-ordinates of the orthocentre of the triangle formed by the lines $y=0$, $37x-36y+37 \times 36=0$ and $64x-63y+64 \times 63=0$ is (a, b) then $a - b - 4535$ is

22. The straight line $L \equiv x + y + 1 = 0$ and $L_1 \equiv x + 2y + 3 = 0$ are intersecting, m is the slope of the straight line L_2 such that L is the bisector of the angle between L_1 and L_2 . The value of m^2 is. Ans : 4

23. The sides of triangle ABC lie on the lines $3x + 4y = 0$, $4x + 3y = 0$ and $x = 3$. Let (h, k) be the center of the circle inscribed in $\triangle ABC$. The value of $(h + k)$ equals.

24. Consider a $\triangle ABC$ whose sides AB, BC , and CA are represented by the straight lines $2x+y=0$, $x+py=q$, and $x - y = 3$, respectively. The point $P(2,3)$ is the orthocenter. The value of $(p + q)/10$ is _____

25. In a square $ABCD$, $G_1=(3,5)$ and $G_2=(5,3)$ are the centroids of $\triangle ABC$ and $\triangle ACD$ then side of the square is (in units)

KEY SHEET (LECTURE SHEET)

EXERCISE-I

LEVEL - I

- 1) 2 2) 1 3) 2 4) 4 5) 2 6) 3 7) 4 8) 3
 9) 3 10) 3 11) 1 12) 1 13) 2 14) 2 15) 1 16) 4

LEVEL - II

- 1) a 2) c 3) c 4) c 5) c 6) d 7) a 8) b
 9) bd 10) ab 11) abcd 12) ab 13) a 14) c 15) c 16) 7
 17) 3 18) 2

EXERCISE-II

LEVEL - I

- 1) 1 2) 2 3) 2 4) 1 5) 2 6) 4 7) 3 8) 2
 9) 3 10) 3 11) 3 12) 3 13) 2 14) 2 15) 2 16) 3
 17) 4 18) 2

LEVEL - II

- 1) b 2) a 3) c 4) b 5) a 6) c 7) b 8) c
 9) a 10) b 11) a 12) d 13) a 14) ac 15) ac 16) abd
 17) ab 18) a 19) c 20) d 21) A-p,q; B-p,q; C-r,s; D-q,r
 22) 1 23) 2 24) 2

EXERCISE-III

LEVEL - I

- 1) 2 2) 3 3) 2 4) 1 5) 3 6) 3 7) 1 8) 2
 9) 2 10) 3 11) 1 12) 2 13) 3 14) 3 15) 1 16) 2
 17) 4 18) 2

LEVEL - II

- 1) a 2) b 3) a 4) a 5) b 6) c 7) b 8) c
 9) b 10) a 11) ab 12) abcd 13) ab 14) bc 15) b 16) c
 17) a 18) a 19) b 20) c 21) 1 22) 4 23) 0 24) 5
 25) 6